

Task 1:

```
UV PICO 5.09 File: myshell.c
#include <stdio.h>
#include <string.h>

int main()
{
    char command[100];
    while (1)
    {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        if (strcmp(command, "exit\n") == 0)
        {
            break;
        }

        printf("You entered: %s", command);
    }

    return 0;
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Out Text UnCut Text Cut Pos To Spell

```
vishalkarnati@VISHALS-MacBook-Air Desktop % cd ~/Desktop
vishalkarnati@VISHALS-MacBook-Air Desktop % nano myshell.c
vishalkarnati@VISHALS-MacBook-Air Desktop % gcc myshell.c -o myshell
vishalkarnati@VISHALS-MacBook-Air Desktop % ./myshell
myshell> hello
You entered: hello
myshell> ls
You entered: ls
myshell> lahari
You entered: lahari
myshell> xit
You entered: xit
myshell> exit
vishalkarnati@VISHALS-MacBook-Air Desktop %
```

Task2:

```
UW PICO 5.09 File: copyfile.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd1, fd2;
    char buffer[100];
    int bytes;

    // Open File1.txt for reading
    fd1 = open("File1.txt", O_RDONLY);

    if (fd1 == -1)
    {
        printf("Error opening File1.txt\n");
        return 1;
    }

    // Open File2.txt for writing
    fd2 = open("File2.txt", O_CREAT | O_WRONLY | O_TRUNC, 0644);

    if (fd2 == -1)
    {
        printf("Error opening File2.txt\n");
        return 1;
    }

    // Copy contents
    while ((bytes = read(fd1, buffer, sizeof(buffer))) > 0)
    {
        write(fd2, buffer, bytes);
    }

    // Close files
    close(fd1);
    close(fd2);

    printf("File copied successfully.\n");

    return 0;
}
```

Get Help WriteOut Read File Prev Pg Out Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

```
vishalkarnati@VISHALS-MacBook-Air Desktop % cd ~/Desktop
vishalkarnati@VISHALS-MacBook-Air Desktop % nano copyfile.c
vishalkarnati@VISHALS-MacBook-Air Desktop % gcc copyfile.c -o copyfile
vishalkarnati@VISHALS-MacBook-Air Desktop % echo "Hello World" > File1.txt
vishalkarnati@VISHALS-MacBook-Air Desktop % cat File1.txt
Hello World
vishalkarnati@VISHALS-MacBook-Air Desktop % ./copyfile
File copied successfully.
vishalkarnati@VISHALS-MacBook-Air Desktop % cat File2.txt
Hello World
vishalkarnati@VISHALS-MacBook-Air Desktop %
```

Task3:

```
UW PICO 5.09 File: task3.c

#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX 100

int main()
{
    char buffer[MAX];
    int index = 0;
    char ch;

    struct termios oldt, newt;
    tcgetattr(STDIN_FILENO, &oldt);

    newt = oldt;
    newt.c_iflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    printf("myshell> ");
    fflush(stdout);

    while (1)
    {
        read(STDIN_FILENO, &ch, 1);

        /* Enter key */
        if (ch == '\n' || ch == '\r')
        {
            buffer[index] = '\0';

            printf("\nYou entered: %s\n", buffer);

            if (strcmp(buffer, "exit") == 0)
            {
                break;
            }

            index = 0;

            printf("myshell> ");
            fflush(stdout);
        }

        /* Backspace key */
        else if (ch == 127 || ch == 8)
        {
            if (index > 0)
            {
                index--;

                printf("\b\b");
                fflush(stdout);
            }
        }
    }

    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    return 0;
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

```
UW PICO 5.09 File: task3.c

        if (ch == '\n' || ch == '\r')
        {
            buffer[index] = '\0';

            printf("\nYou entered: %s\n", buffer);

            if (strcmp(buffer, "exit") == 0)
            {
                break;
            }

            index = 0;

            printf("myshell> ");
            fflush(stdout);
        }

        /* Backspace key */
        else if (ch == 127 || ch == 8)
        {
            if (index > 0)
            {
                index--;

                printf("\b\b");
                fflush(stdout);
            }
        }

        /* Normal character */
        else
        {
            if (index < MAX - 1)
            {
                buffer[index] = ch;
                index++;

                write(STDOUT_FILENO, &ch, 1);
            }
        }

        tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    return 0;
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

```

vishalkarnati@VISHALS-MacBook-Air ~ % nano task3.c
vishalkarnati@VISHALS-MacBook-Air ~ % gcc task3.c -o task3
vishalkarnati@VISHALS-MacBook-Air ~ % ./task3
myshell> hello
You entered: hello
myshell> hello world
You entered: hello world
myshell> hellox
You entered: hellox
myshell> exit
You entered: exit
vishalkarnati@VISHALS-MacBook-Air ~ %

```

Task4:

```

UW PICO 5.09
File: task4.c

#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

#define MAX_INPUT 100
#define MAX_HISTORY 10

char history[MAX_HISTORY][MAX_INPUT];

int history_count = 0;
int history_index = 0;

void clear_line()
{
    printf("\033[2K\r");
    printf("myshell> ");
}

int main()
{
    char buffer[MAX_INPUT];
    int index;
    char ch;

    struct termios oldt, newt;

    /* Save original terminal settings */
    tcgetattr(STDIN_FILENO, &oldt);

    /* Create new terminal settings */
    newt = oldt;

    newt.c_lflag &= ~(ICANON | ECHO);

    /* Apply new settings */
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    while (1)
    {
        printf("myshell> ");
        fflush(stdout);

        index = 0;
        buffer[0] = '\0';

        while (1)
        {
            read(STDIN_FILENO, &ch, 1);

            /* ENTER */
            if (ch == '\n' || ch == '\r')
            {
                buffer[index] = '\0';
                printf("\n");
                break;
            }
        }
    }
}

```

 Get Help
 Exit
  WriteOut
 Justify
  Read File
 Where is
  Prev Pg
 Next Pg
  Cut Text
 UnCut Text
  Cur Pos
 To Spell

```
UW PICO 5.09 File: task4.c

/* Create new terminal settings */
newt = oldt;

newt.c_lflag &= ~(ICANON | ECHO);

/* Apply new settings */
tcsetattr(STDIN_FILENO, TCSANOW, &newt);

while (1)
{
    printf("myshell> ");
    fflush(stdout);

    index = 0;
    buffer[0] = '\0';

    while (1)
    {
        read(STDIN_FILENO, &ch, 1);

        /* ENTER */
        if (ch == '\n' || ch == '\r')
        {
            buffer[index] = '\0';
            printf("\n");
            break;
        }

        /* BACKSPACE */
        else if (ch == 127 || ch == 8)
        {
            if (index > 0)
            {
                index--;
                printf("\b \b");
                fflush(stdout);
            }
        }

        /* ESCAPE SEQUENCE */
        else if (ch == 27)
        {
            char seq1;
            char seq2;

            read(STDIN_FILENO, &seq1, 1);
            read(STDIN_FILENO, &seq2, 1);

            /* UP ARROW */
            if (seq1 == '[' && seq2 == 'A')
            {
                if (history_count > 0 && history_index > 0)
                {
                    history_index--;
                }
            }
        }
    }
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

```
UW PICO 5.09 File: task4.c

/* BACKSPACE */
else if (ch == 127 || ch == 8)
{
    if (index > 0)
    {
        index--;
        printf("\b \b");
        fflush(stdout);
    }
}

/* ESCAPE SEQUENCE */
else if (ch == 27)
{
    char seq1;
    char seq2;

    read(STDIN_FILENO, &seq1, 1);
    read(STDIN_FILENO, &seq2, 1);

    /* UP ARROW */
    if (seq1 == '[' && seq2 == 'A')
    {
        if (history_count > 0 && history_index > 0)
        {
            history_index--;
            strcpy(buffer, history[history_index]);
            index = strlen(buffer);
            clear_line();
            printf("Ms", buffer);
            fflush(stdout);
        }
    }

    /* DOWN ARROW */
    else if (seq1 == '[' && seq2 == 'B')
    {
        if (history_index < history_count - 1)
        {
            history_index++;
            strcpy(buffer, history[history_index]);
            index = strlen(buffer);
            clear_line();
            printf("Ms", buffer);
            fflush(stdout);
        }
        else
        {
        }
    }
}
```

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell

```
UW PICO 5.09 File: task4.c

    index = strlen(buffer);
    clear_line();
    printf("%s", buffer);
    fflush(stdout);
}
}

/* DOWN ARROW */
else if (seq1 == '[' && seq2 == 'B')
{
    if (history_index < history_count - 1)
    {
        history_index++;
        strcpy(buffer, history[history_index]);
        index = strlen(buffer);
        clear_line();
        printf("%s", buffer);
        fflush(stdout);
    }
    else
    {
        history_index = history_count;
        buffer[0] = '\0';
        index = 0;
        clear_line();
        fflush(stdout);
    }
}

/* NORMAL CHARACTER */
else
{
    if (index < MAX_INPUT - 1)
    {
        buffer[index] = ch;
        index++;
        write(STDOUT_FILENO, &ch, 1);
    }
}

/* EXIT */
if (strcmp(buffer, "exit") == 0)
{
    Get Help
    Exit
    WriteOut
    Justify
    Read File
    Where is
    Prev Pg
    Next Pg
    Cut Text
    UnCut Text
    Cur Pos
    To Spell
}
```

```
UW PICO 5.09 File: task4.c

    history_index = history_count;
    buffer[0] = '\0';
    index = 0;
    clear_line();
    fflush(stdout);
}
}

/* NORMAL CHARACTER */
else
{
    if (index < MAX_INPUT - 1)
    {
        buffer[index] = ch;
        index++;
        write(STDOUT_FILENO, &ch, 1);
    }
}

/* EXIT */
if (strcmp(buffer, "exit") == 0)
{
    break;
}

/* STORE COMMAND IN HISTORY */
if (strlen(buffer) > 0)
{
    if (history_count < MAX_HISTORY)
    {
        strcpy(history[history_count], buffer);
        history_count++;
    }
    history_index = history_count;
}

printf("You entered: %s\n", buffer);
}

/* Restore original terminal settings */
tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

return 0;
}
```

```
vishalkarnati@VISHALS-MacBook-Air Desktop % nano task4.c
vishalkarnati@VISHALS-MacBook-Air Desktop % gcc task4.c -o task4
vishalkarnati@VISHALS-MacBook-Air Desktop % ./task4
myshell> hello
You entered: hello
myshell> apple
You entered: apple
myshell> banana
You entered: banana
myshell> hello world
You entered: hello world
myshell> exit
vishalkarnati@VISHALS-MacBook-Air Desktop %
```

Task 5:

```
UW PICO 5.09 File: task5.c

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct Node
{
    char *command;
    struct Node *next;
} Node;

Node *head = NULL;

/* Add a command to the linked list */
void add_history(const char *command)
{
    Node *new_node;
    new_node = malloc(sizeof(Node));

    if (new_node == NULL)
    {
        perror("malloc failed");
        exit(1);
    }

    new_node->command = malloc(strlen(command) + 1);

    if (new_node->command == NULL)
    {
        perror("malloc failed");
        free(new_node);
        exit(1);
    }

    strcpy(new_node->command, command);
    new_node->next = NULL;

    if (head == NULL)
    {
        head = new_node;
    }
    else
    {
        Node *current = head;

        while (current->next != NULL)
        {
            current = current->next;
        }

        current->next = new_node;
    }
}

/* Display all stored commands */
void display_history()
{
    Node *current = head;
    while (current != NULL)
    {
        printf("%s\n", current->command);
        current = current->next;
    }
}
```

```
UW PICO 5.09 File: task5.c

    free(new_node);
    exit(1);
}

strcpy(new_node->command, command);

new_node->next = NULL;

if (head == NULL)
{
    head = new_node;
}
else
{
    Node *current = head;

    while (current->next != NULL)
    {
        current = current->next;
    }

    current->next = new_node;
}

/* Display all stored commands */
void display_history()
{
    Node *current = head;
    int number = 1;

    while (current != NULL)
    {
        printf("%d %s\n", number, current->command);
        current = current->next;
        number++;
    }
}

/* Free the complete linked list */
void free_history()
{
    Node *current = head;

    while (current != NULL)
    {
        Node *temp = current;
        current = current->next;
        free(temp->command);
        free(temp);
    }

    head = NULL;
}

int main()
{
    char command[100];

    while (1)
    {
        printf("Enter a command: ");
        fgets(command, sizeof(command), stdin);

        if (command[0] == '\n')
            continue;

        add_history(command);

        display_history();

        printf("Press Ctrl+C to exit\n");
    }

    free_history();
}
```

```
UW PICO 5.09 File: task5.c

while (current != NULL)
{
    printf("%d %s\n", number, current->command);
    current = current->next;
    number++;
}
}

/* Free the complete linked list */
void free_history()
{
    Node *current = head;
    while (current != NULL)
    {
        Node *temp = current;
        current = current->next;
        free(temp->command);
        free(temp);
    }
    head = NULL;
}

int main()
{
    char *buffer = NULL;
    size_t buffer_size = 0;
    ssize_t length;

    while (1)
    {
        printf("myshell> ");
        fflush(stdout);

        length = getline(&buffer, &buffer_size, stdin);

        if (length == -1)
        {
            break;
        }

        if (length > 0 && buffer[length - 1] == '\n')
        {
            buffer[length - 1] = '\0';
        }

        if (strcmp(buffer, "exit") == 0)
        {
            break;
        }

        if (strcmp(buffer, "history") == 0)
        {
            display_history();
        }
    }
}

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell
```

```
UW PICO 5.09 File: task5.c

char *buffer = NULL;
size_t buffer_size = 0;
ssize_t length;

while (1)
{
    printf("myshell> ");
    fflush(stdout);

    length = getline(&buffer, &buffer_size, stdin);

    if (length == -1)
    {
        break;
    }

    if (length > 0 && buffer[length - 1] == '\n')
    {
        buffer[length - 1] = '\0';
    }

    if (strcmp(buffer, "exit") == 0)
    {
        break;
    }

    if (strcmp(buffer, "history") == 0)
    {
        display_history();
        continue;
    }

    if (strlen(buffer) > 0)
    {
        add_history(buffer);
    }

    printf("You entered: %s\n", buffer);

    free(buffer);
    free_history();
}

return 0;
}

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell
```

```
vishalkarnati@VISHALS-MacBook-Air os % nano task5.c
vishalkarnati@VISHALS-MacBook-Air os % gcc task5.c -o task5
vishalkarnati@VISHALS-MacBook-Air os % ./task5
myshell> hello
You entered: hello
myshell> apple
You entered: apple
myshell> banana
You entered: banana
myshell> history
1 hello
2 apple
3 banana
myshell> exit
vishalkarnati@VISHALS-MacBook-Air os %
```